Ilyes Saddi

Graduation Internship (PFE) – Backend & AI Engineer – 6 months starting February 2027

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SUMMARY

Final-year **Computer Science Engineering** student at ENSI (**Master's-level, Bac+5 equivalent**), specializing in **backend development** , as well as applied AI/ML**,** including LLMs and deep learning. Seeking a **six-month graduation internship** starting in **February 2027** to design and develop scalable backend systems and AI-powered solutions.

EDUCATION

09/2024 – Present **National School for Computer Science** Manouba, Tunisia
Computer Science Engineering (Degree)

09/2022 – 07/2024 **Preparatory Institute for Engineering Studies of Monastir** Monastir
Math-Physics-Chemistry-MSI-IT

PROFESSIONAL EXPERIENCE

08/2026 – 09/2026 **ACOBA — software engineer Internship** Monastir
AI / Test Automation Developer Internship — Remote
- Built a non-regression test automation tool for the Cam2Drive web interface, using an LLM (Amazon Bedrock) to generate Playwright tests from user scenarios.
- Developed a code retriever that maps a scenario to the relevant files and functions, providing context to the LLM prompt.

01/2025 – 01/2026 **Project Lead | ROBOTIQUE ENSI Association** Manouba
Project Manager
Led association projects, responsible for planning, coordination, and successful delivery of technical initiatives .

07/2025 – 08/2025 **Device Speak - Internship** Djerba
Full-Stack Developer
Develop a **full stack** website using Graphql api
- Developed a real-time web application using **React (Vite)**, **Node.js**, and **MongoDB**.
- Integrated **GraphQL APIs** to build a dynamic dashboard with efficient **real-time data handling**.

07/2024 – 08/2024 **Solution BE - Internship** Djerba
- Designed underground fiber routes using **GPS** for clients in **Germany**.
- Managed technical equipment and machinery to ensure 100% operational success for client networks.

PROJECTS

09/2026 – Present **Energy Price Guardian — ASP.NET Core Web API**
- Developing a backend application using C#, ASP.NET Core Web API, and Entity Framework Core.
- Designing **RESTful API** endpoints and structuring backend services using **dependency injection** and **middleware**.
- Exploring electricity price data integration to identify lower-cost hours for household energy consumption.

C#, ASP.NET Core Web API, Entity Framework Core, **MySQL**, **REST APIs**, Dependency Injection.

01/2026 – 06/2026 **Chest X-ray Disease Detection System | PCD**
•Built a classification pipeline for multi-label chest X-ray disease
- Fine-tuned a MedGemma model to generate reports
- Integrated CheXbert to validate labels

Python, PyTorch, **PEFT (LoRA),** MedGemma, CheXbert, Computer Vision, Deep Learning, Multimodal Learning, **Medical Image Preprocessing**, **Feature Fusion**, Multi-Label Classification**.**

Cam2Drive Test non regression
Development of a non-regression **test automation** solution for the Cam2Drive interface
- Develop a **Code Retriever** to identify relevant files and functions within the project after knowing the senario
- Automatically generate **Playwright** tests based on code context and user scenarios.
- Integrate a **LLM** for test generation, with a planned evolution toward **Amazon Bedrock**.

09/2025 – 2025 **Machine Learning Mini Project:**
Voting Participation Prediction using demographic data
Supervised classification, data preprocessing, model training & accuracy evaluation

01/2025 – 06/2025 **Secure Entry | embedded System project**
- Developed an automated security system using **ESP32-CAM** microcontrollers and a **Raspberry**, and manage data **through** integrated memory. **used technologies :** Python, OpenCV (Computer Vision) / MQTT, HTTP, IoT Architecture/ ESP32, Raspberry Pi .

10/2024 – 12/2024 **Hotel Management System | C Developer**
- Built a management platform using **Linked Lists**, **Queues**, and **File Handling** to automate room reservations and maintenance scheduling.
- Optimized data retrieval and storage processes to manage room availability and billing calculations in real-time.

SKILLS

- **Programming Languages:** C#, SQL , Python, C/C++
- **Frameworks & Libraries:** ASP.net , React (Vite)
- **Databases:** MongoDB (NoSQL), MySQL, PostgreSQL.
- **Developer Tools & Technologies:** GraphQL, Git/GitHub, VS Code.
- **Software Development Concepts:** Object-Oriented Programming (OOP), SOLID Principles, Design Patterns, Data Structures and Algorithms
- **Machine Learning & Deep Learning:**Supervised Learning, Classification, Neural Networks, Convolutional neural network,Deep learning , LSTM , RNN, Hugging Face Transformers,prompt engineering

ORGANISATIONS

2022 **Tunisian Scouts (Les Scouts Tunisiens)** Ajim Djerba Tunisia
Patrol Leader
- Managed and mentored a group of scouts, coordinating weekly activities and outdoor expeditions.

AWARDS

24/11/2025 **Second Place in Machine Learning Competition**
Orbyx ENSI Club
- Developed predictive models using **Python** (**Pandas**, **Scikit-learn**) to solve complex data challenges.

27/04/2025 **Second Place Line Follower Competition**
Robotek ISET Rades
- Designed and optimized a high-speed autonomous robot using **C++** and **PID control** algorithms to navigate complex tracks.